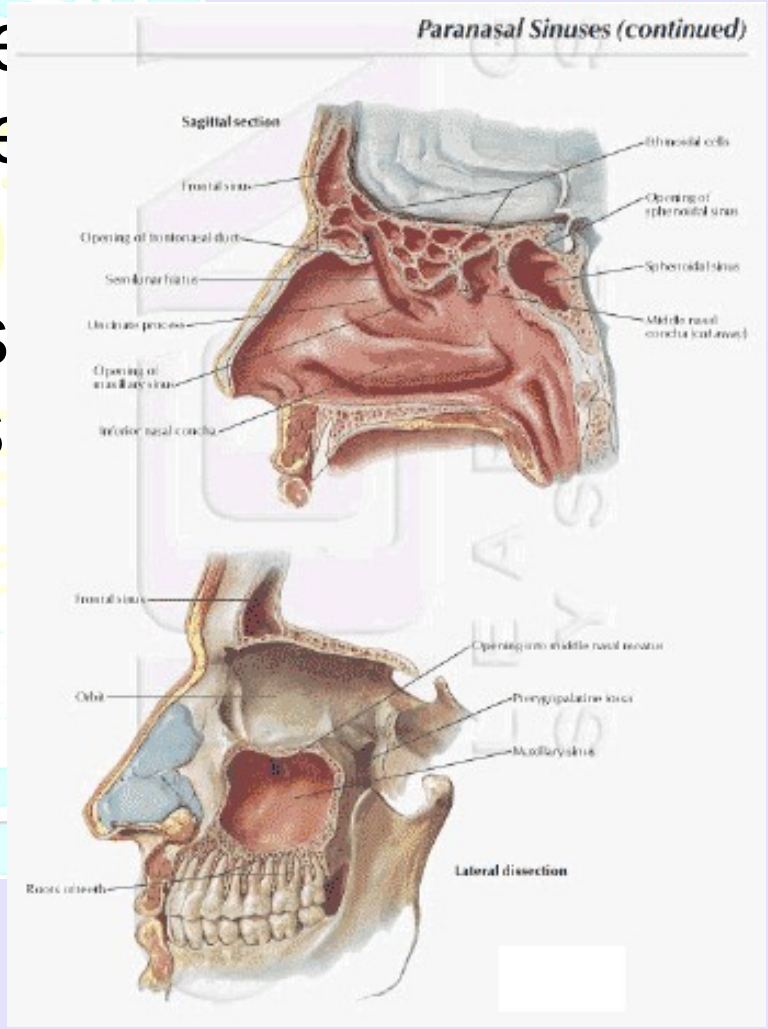
Paranasal sinuses

Learning objectives

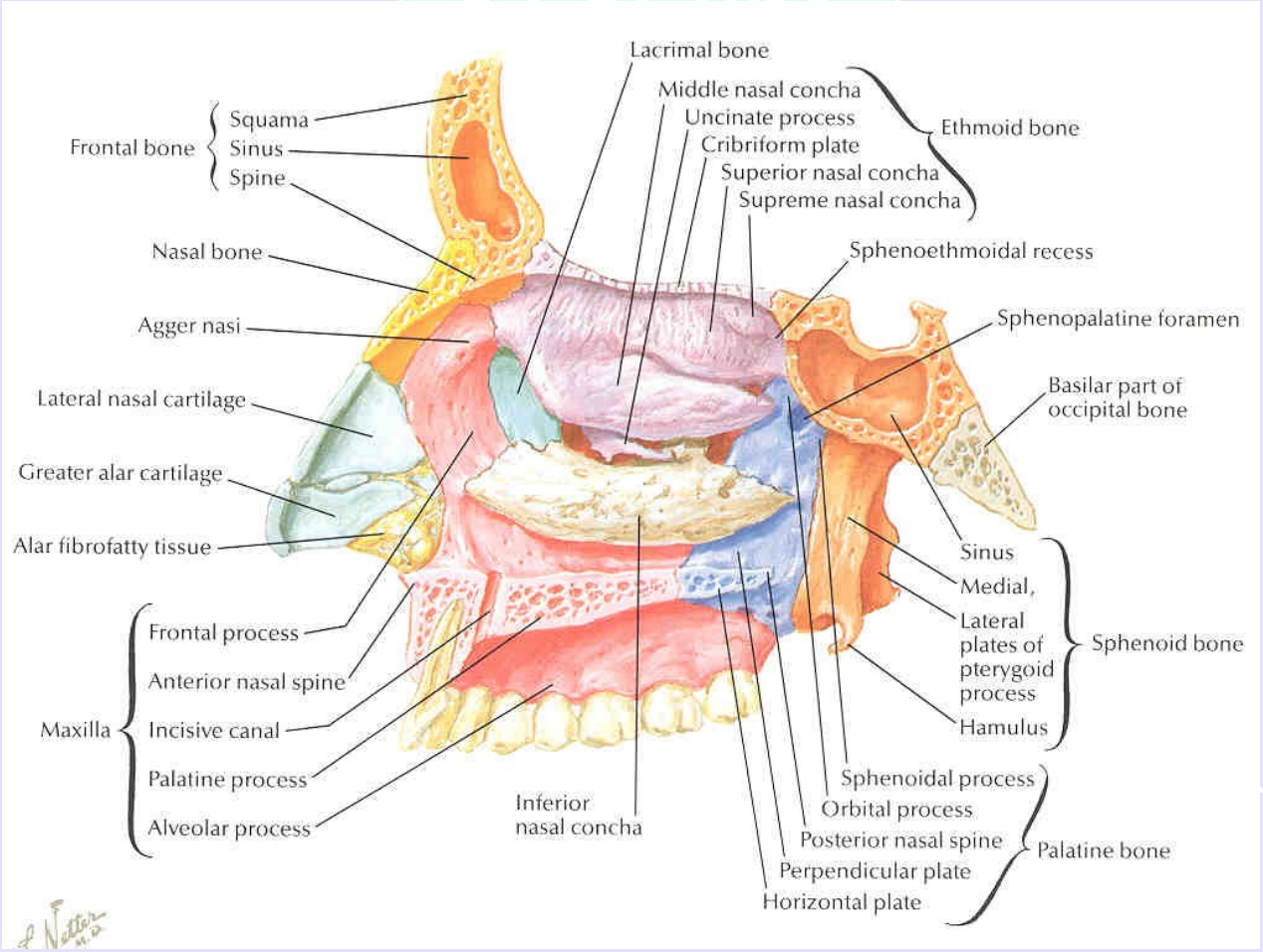
- At the end of the lecture the student should be able to know:
- Definition of paranasal sinuses
- Names of paranasal sinuses
- Functions of paranasal sinuses.
- Radiographic Protocols for sinuses
- Diseases of sinuses.

Paranasal sinuses

- **Paranasal sinuses** are
communicating with the
within the bones of the
- Divided into subgroups
according to the bones
sinuses lie

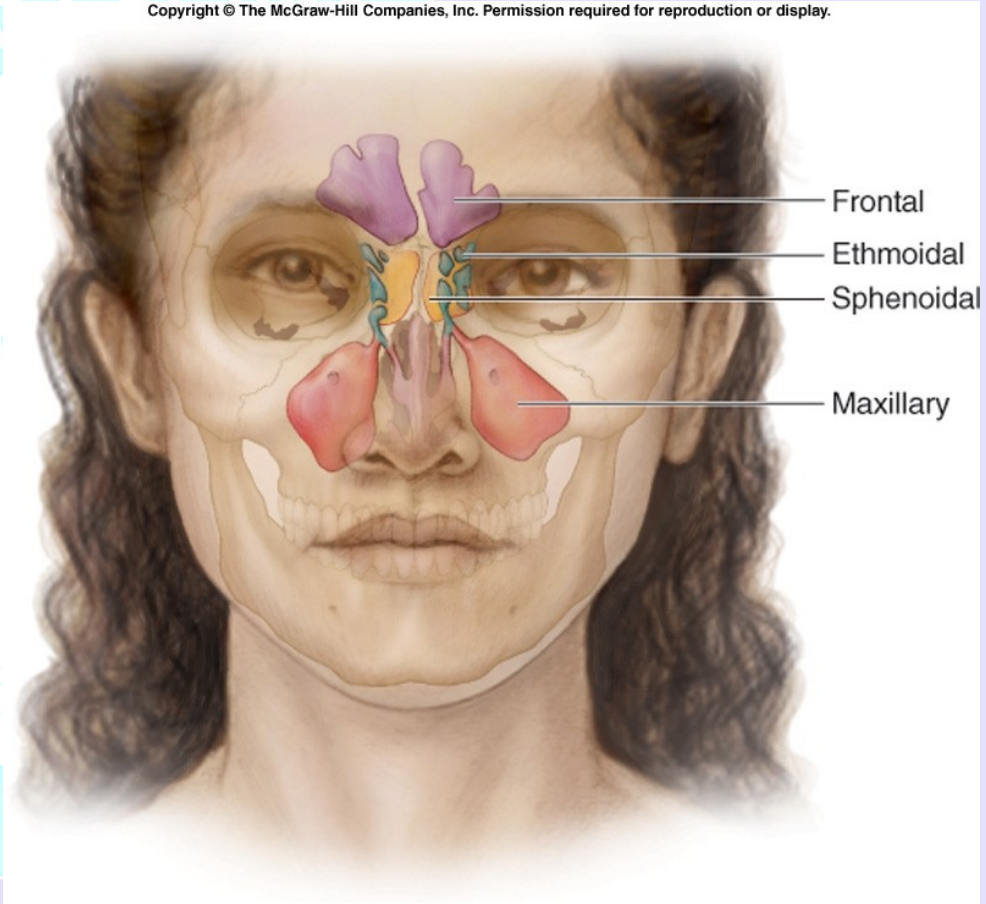


Bony structure of nose



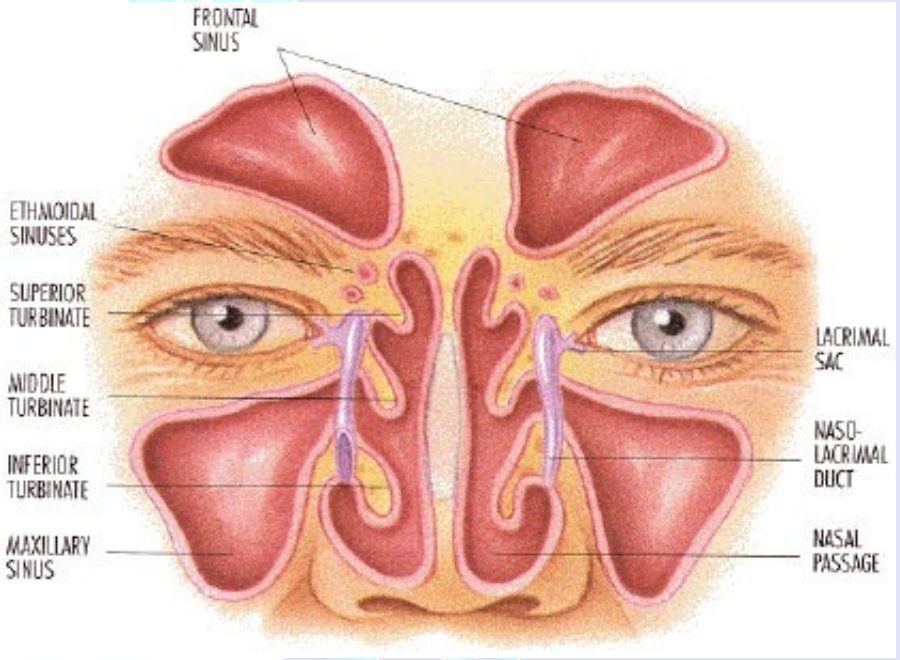
Names of sinuses

- Maxillary
- Frontal
- Ethmoid
- Sphenoid



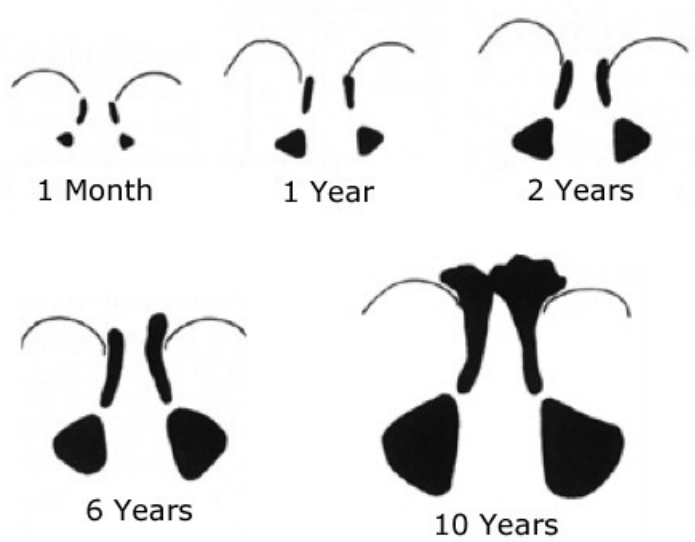
Maxillary sinus

- Present at birth
- Largest sinus
- Found in body of maxilla
- Has 3 recesses:
  - Apex of maxillary sinus extends into zygomatic bone.
  - Base forms the inferior part of lateral wall of nasal cavity
  - Roof formed by the floor of the orbit.
  - Floor formed by the alveolar part of maxilla.
    - Roots of maxillary teeth (2

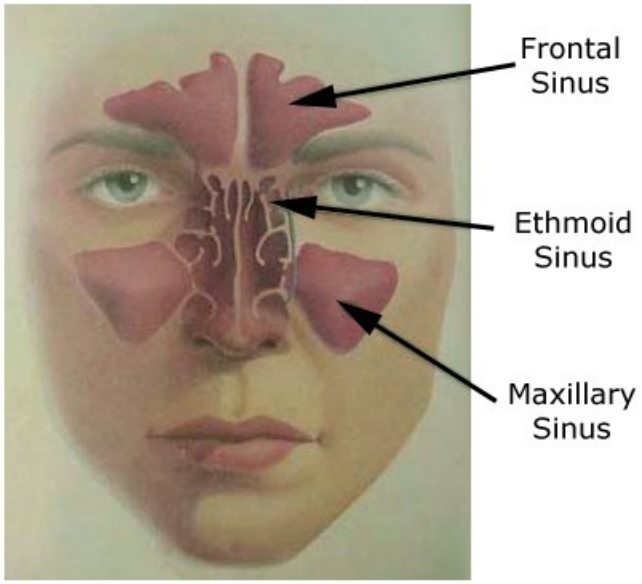


Development of sinuses

- Aerated at birth
  - **Maxillary sinuses**
- Age 6-7
  - **Frontal /sphenoidal sinuses**
- Puberty- approx 17-18 yrs
  - **Ethmoid**



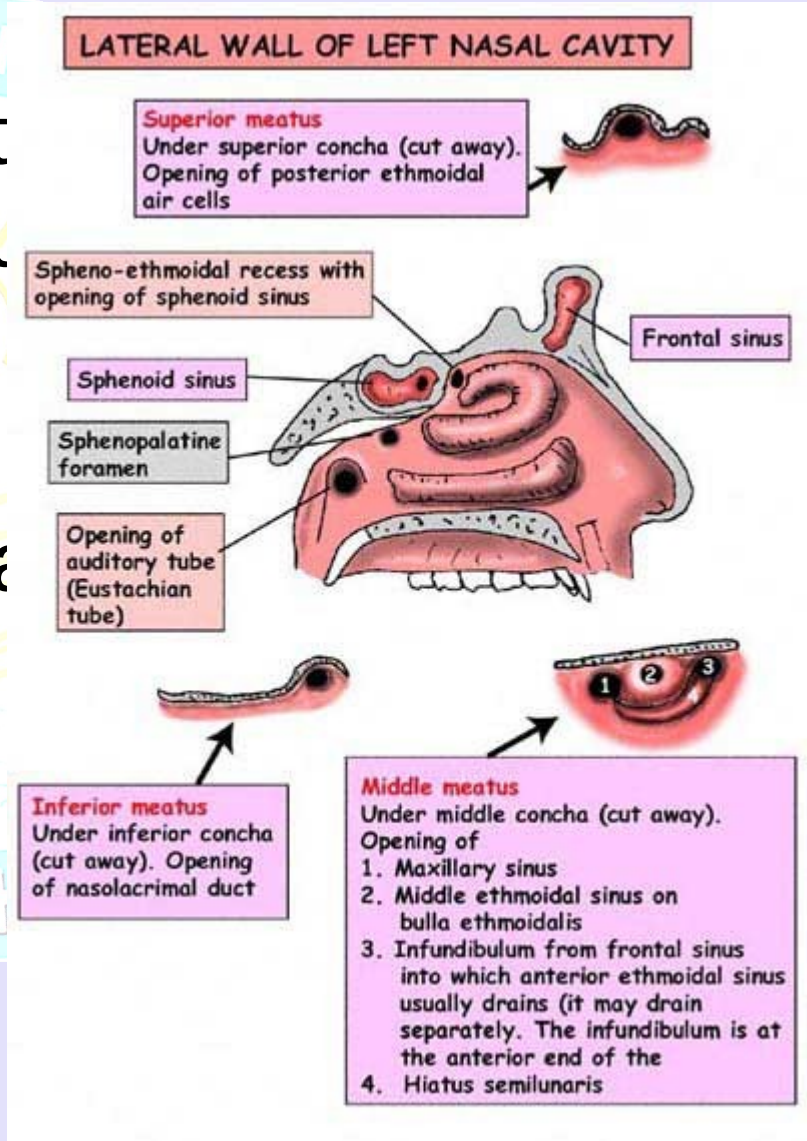
Pediatric Sinus Anatomy



Adult Sinus Anatomy
(Sphenoid Sinus not Shown)

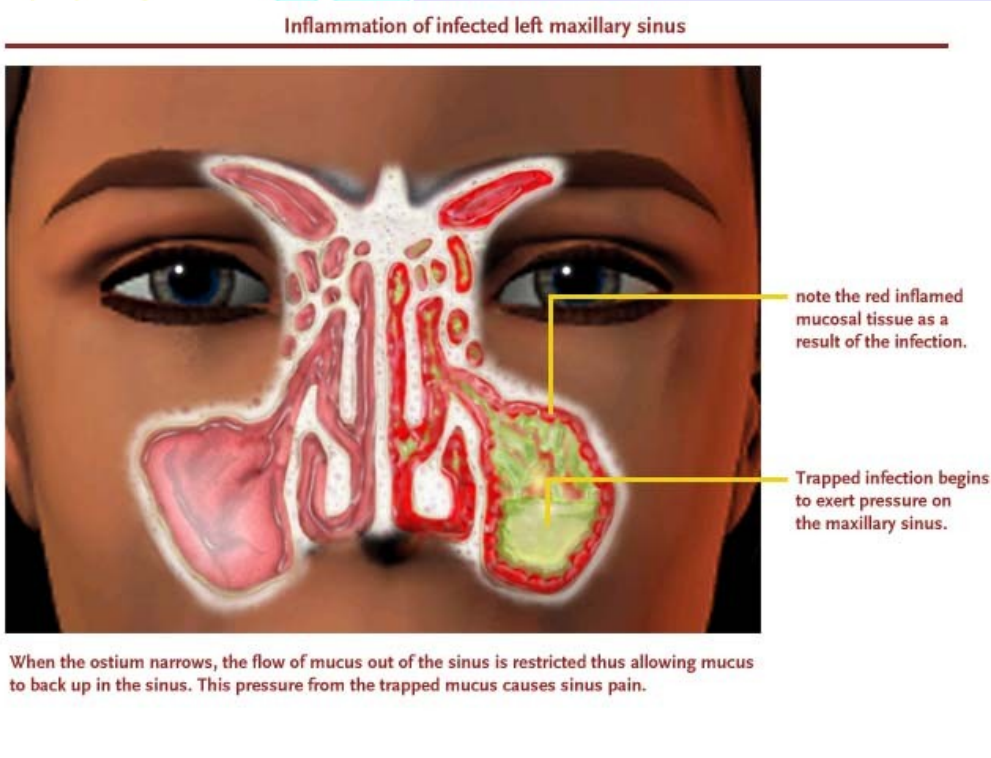
Ostium of maxillary sinus

- The sinus communicat
  opening into the semilu
  lateral nasal wall.
- The sinus is lined with
  with cilia that beat towa



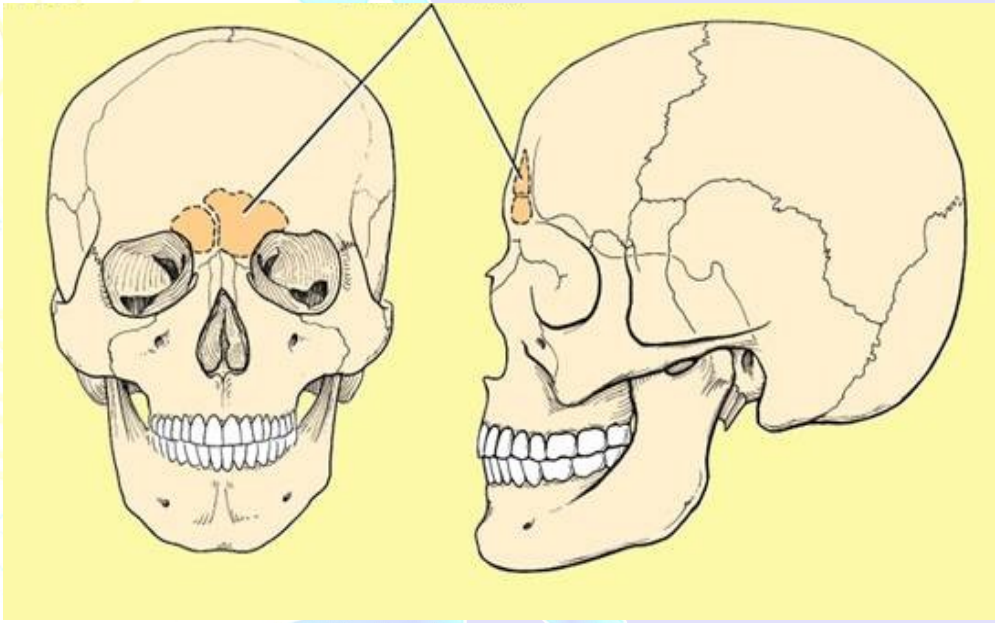
Maxillary sinusitis

- Inflammation of maxillary sinus
- Maxillary sinusitis is common due to the close anatomical relation of the frontal sinus, anterior ethmoidal air sinus and the maxillary teeth, allowing for easy spread of infection.
- Drainage orifice lies near the roof of the sinus, and so the maxillary sinus does not drain at all well, and infection develops more easily.
- Treatment
- Broad spectrum antibiotic



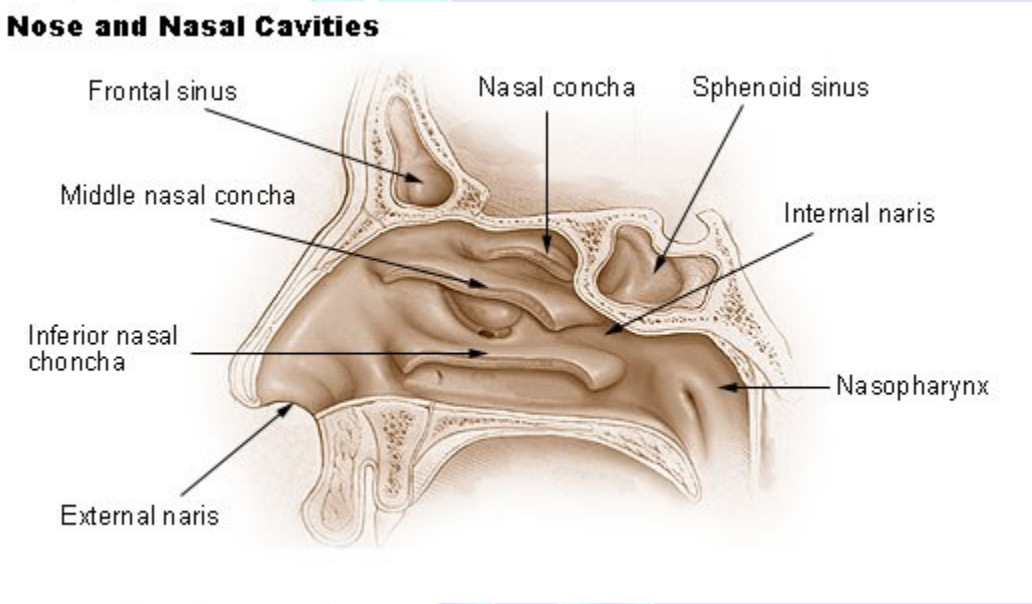
Frontal sinus

- Second largest sinuses
- Number varies
- Situated behind the superciliary arches,
- Absent at birth, reaching their full size after puberty.
- Frontal sinuses are rarely symmetrical and the septum between them frequently deviates to one or other side of the middle line



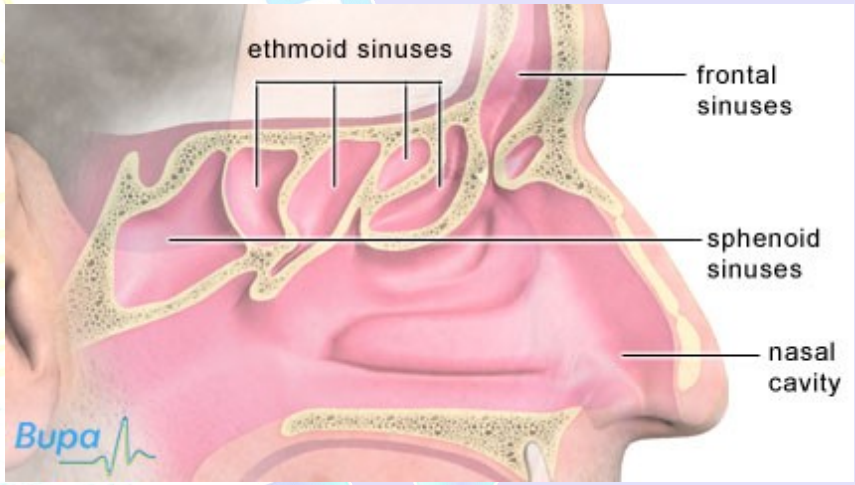
Opening of frontal sinus

- Opens into the anterior part of the corresponding middle meatus of the nose through the frontonasal duct.
- Each frontal sinus opens into the anterior part of the middle meatus of the nose through the frontonasal duct.
- Which opens into the hiatus semilunaris
- The **hiatus semilunaris** is a halfmoon-shaped groove in the lateral wall of the nasal cavity.

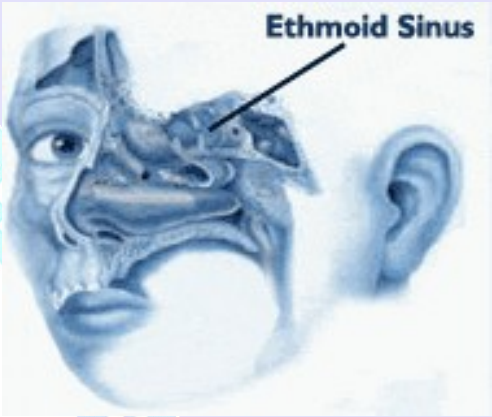


Ethmoid sinuses

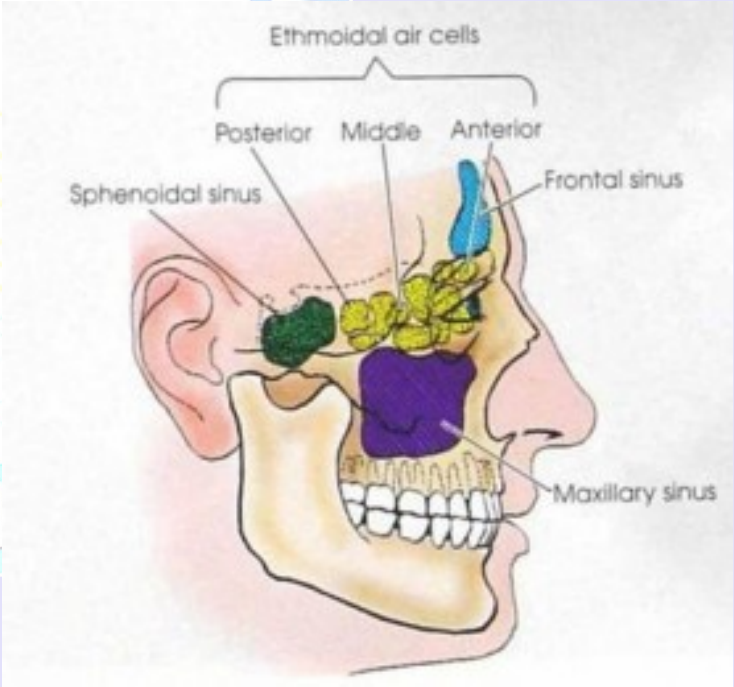
- Collective name for the **ethmoidal air cells**.
- Each ethmoid sinus is an air-space enclosed within the ethmoid bone. The ethmoidal air cells consist of numerous thin-walled cavities situated in the ethmoidal labyrinth and completed by the frontal, maxilla, lacrimal, sphenoidal, and palatine bones.
- They lie between the orbits, above the nasal cavity and are separated from these cavities by thin bony laminae.



Ethmoid sinuses opening

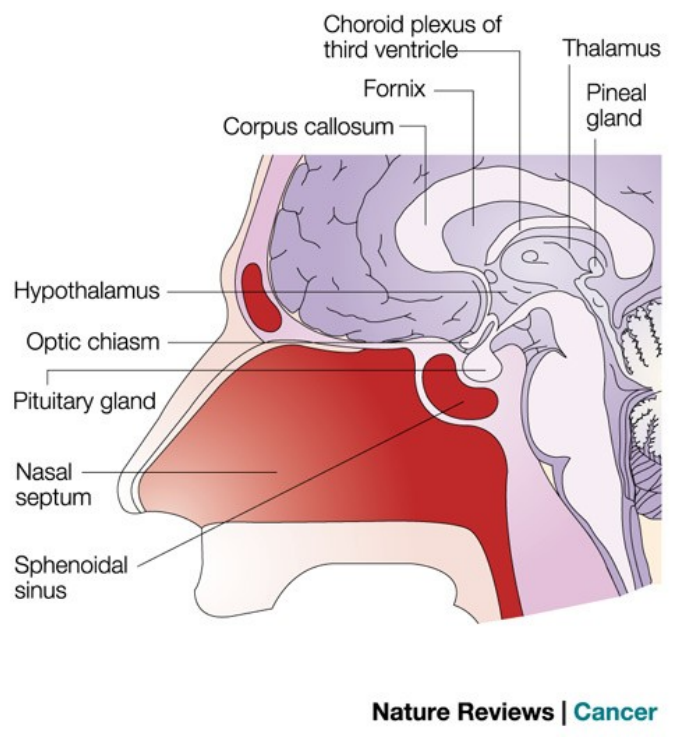


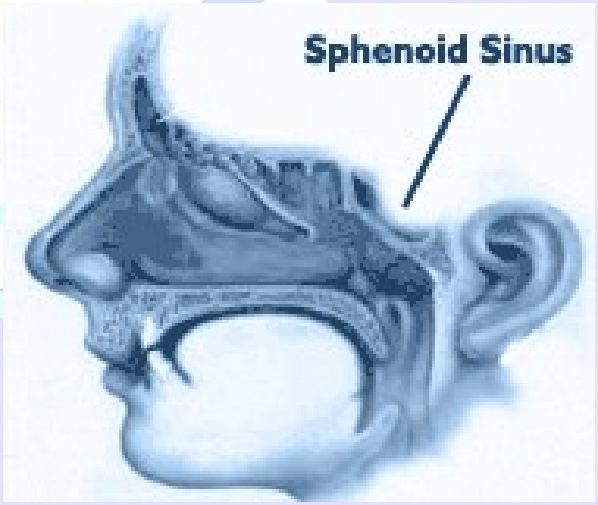
- Anterior, middle, and posterior ethmoid sinuses.
- P**osterior** drains into the superior meatus under cover of the superior nasal concha; sometimes one or more opens into the sphenoidal sinus.
- **Middle** drains into the middle meatus of the nose on or above the bulla ethmoidalis.
- **Anterior** drains into the middle meatus of the nose by way of the infundibulum.



Sphenoid sinus

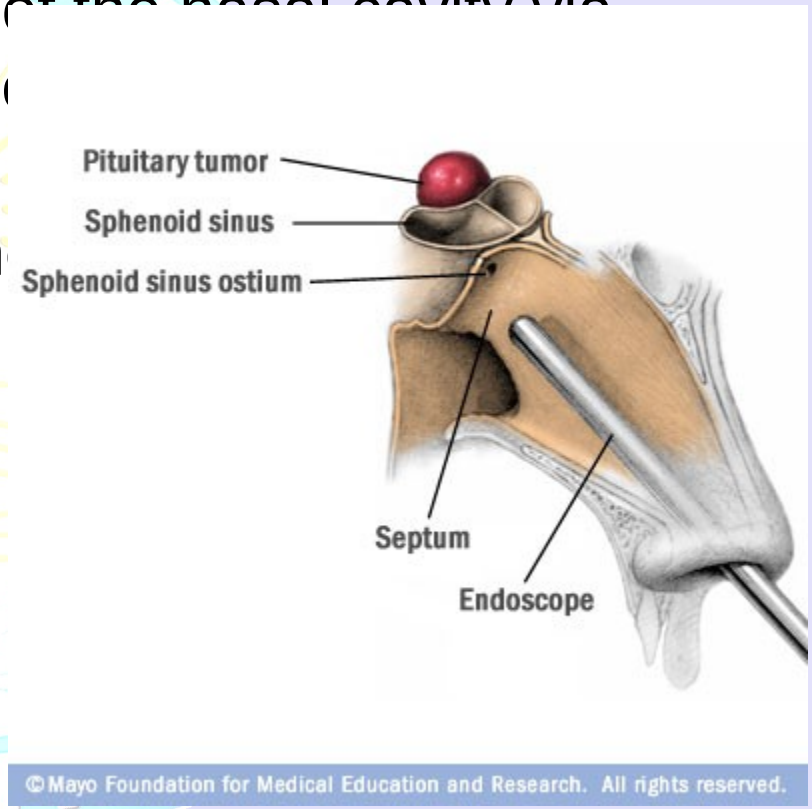
- They are present as minute cavities at birth, but their main development takes place after puberty.
- Present within the body of sphenoid.
- They are rarely symmetrical due to the lateral displacement of the intervening septum.
- When exceptionally large they may extend into the basilar part of the occipital bone.
- Pituitary gland can be approached through nose bypassing the sphenoid sinus.





Opening of sphenoid sinus

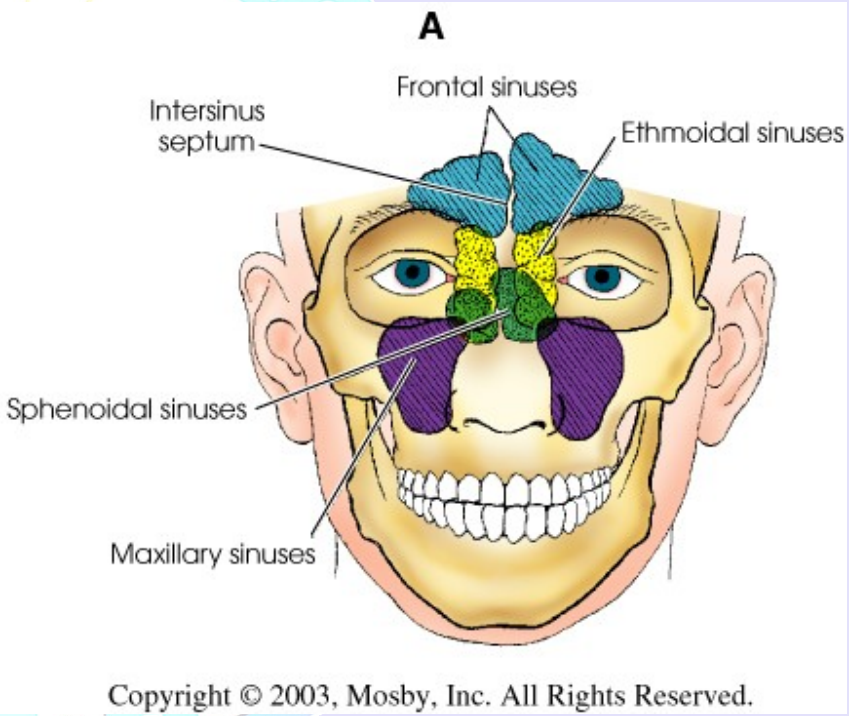
- Each sinus opens into the roof of the nasal cavity via apertures on the posterior wall of recess.
- A potential complication of sph cavernous sinus thrombosis.



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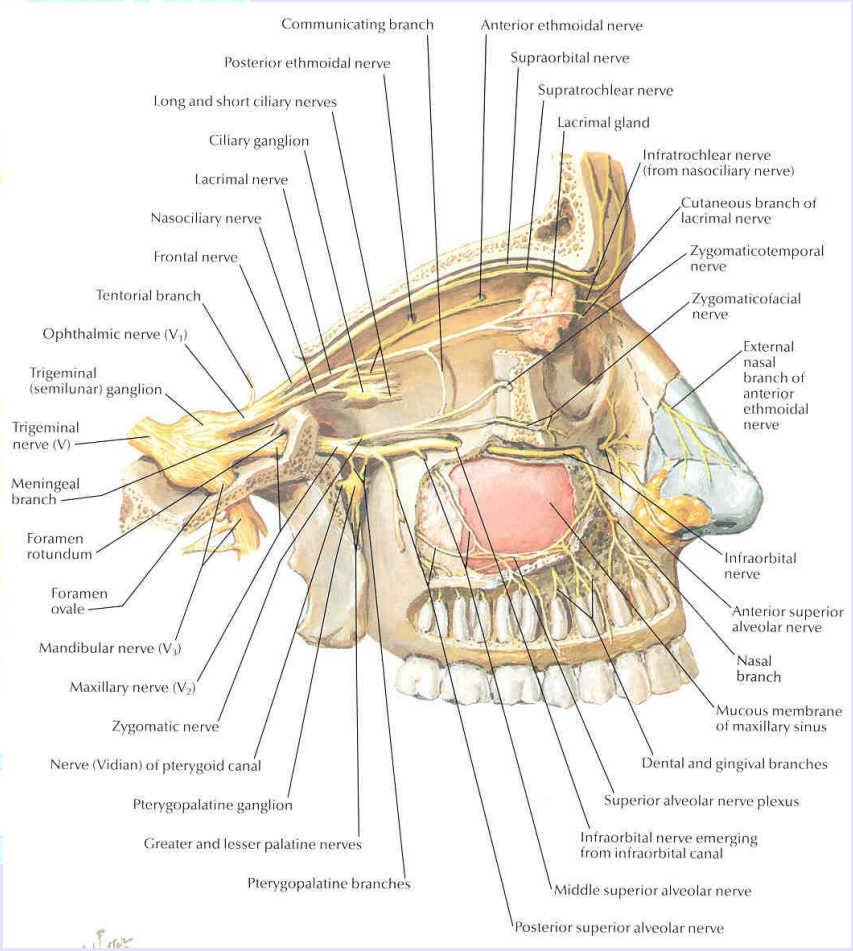
Function of paranasal sinuses

- Resonating chamber for voice
- Decrease weight of skull
- Warm & moisten air
- Shock absorbers
- Immune system



Innervation

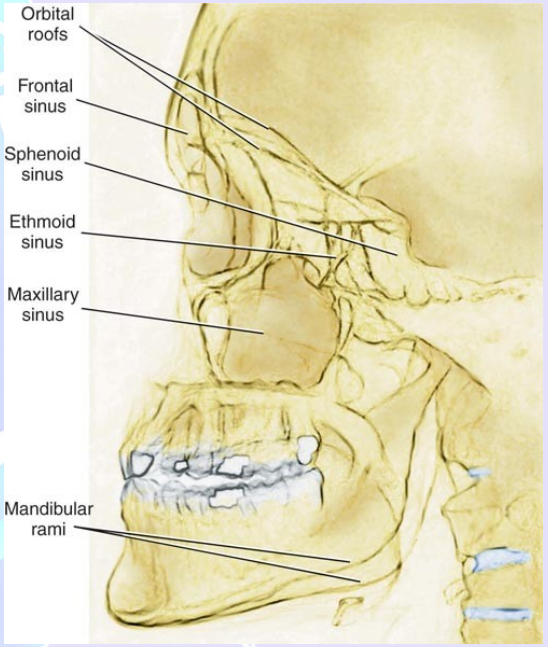
- Maxillary
  - Superior alveolar nerve branch of maxillary nerve.



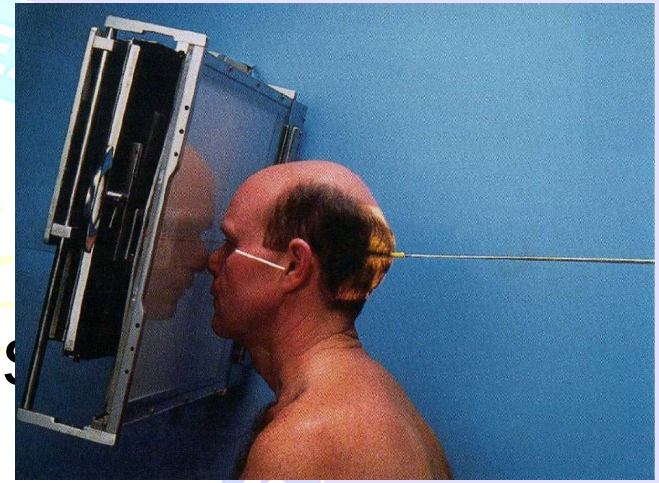
Radiology

- Lateral Radiograph
- All 4 sinuses
  - Sphenoid of primary interest

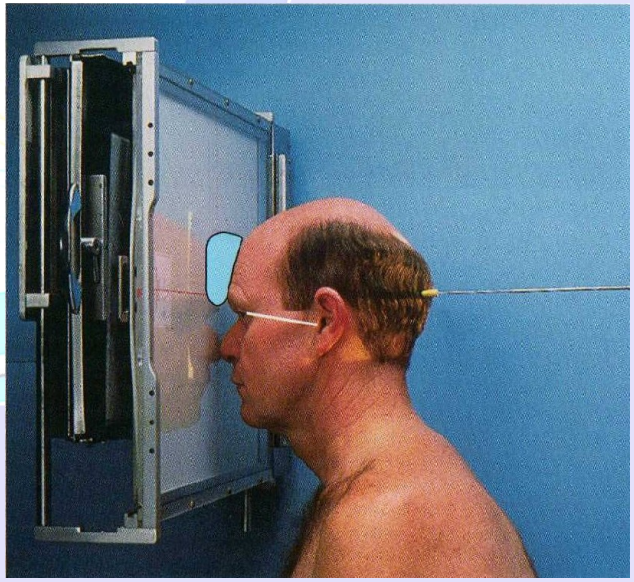




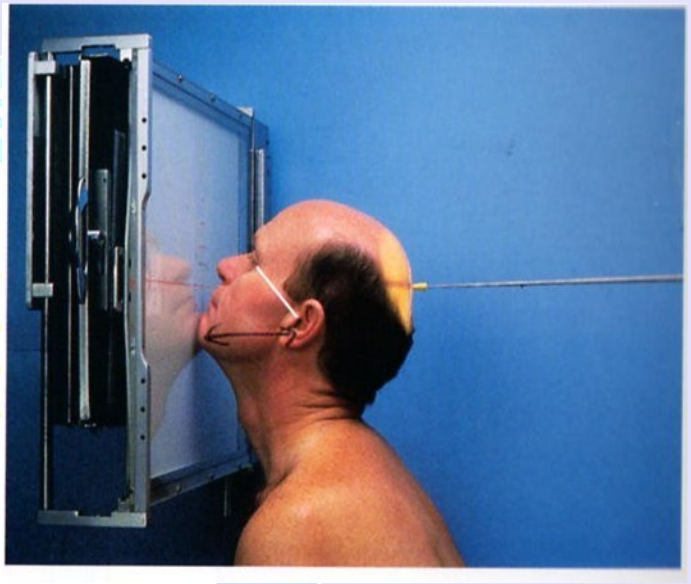
- (PA) caldwell.
- Angled and vertical.
- Frontal &ethmoid air cells



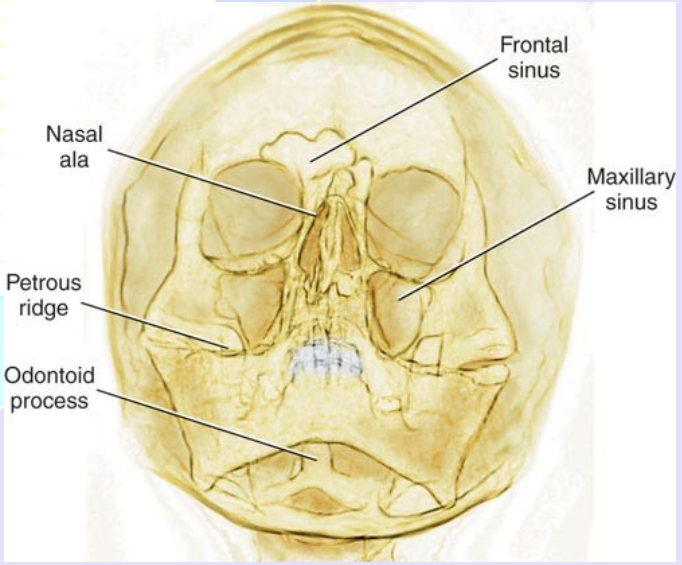




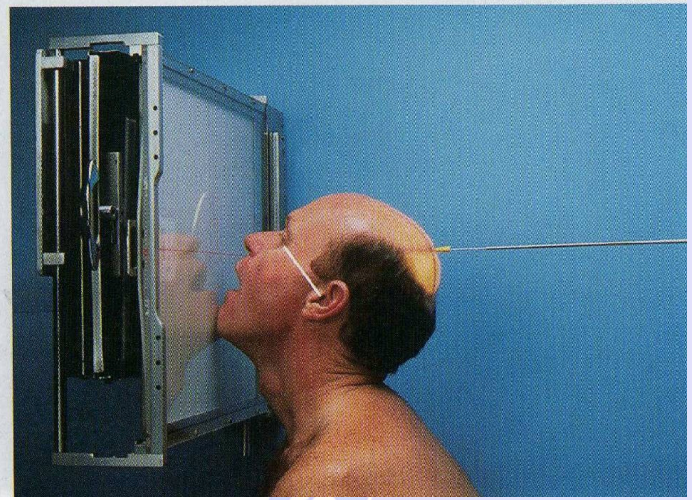
- Waters view.
- For maxillary sinus

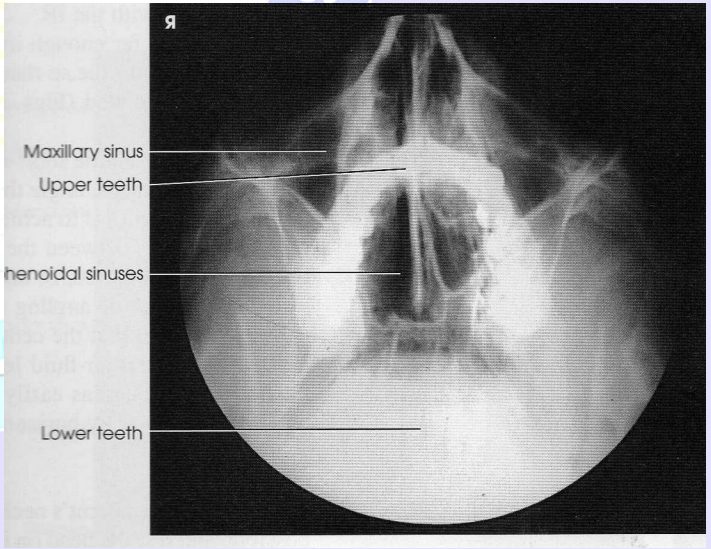




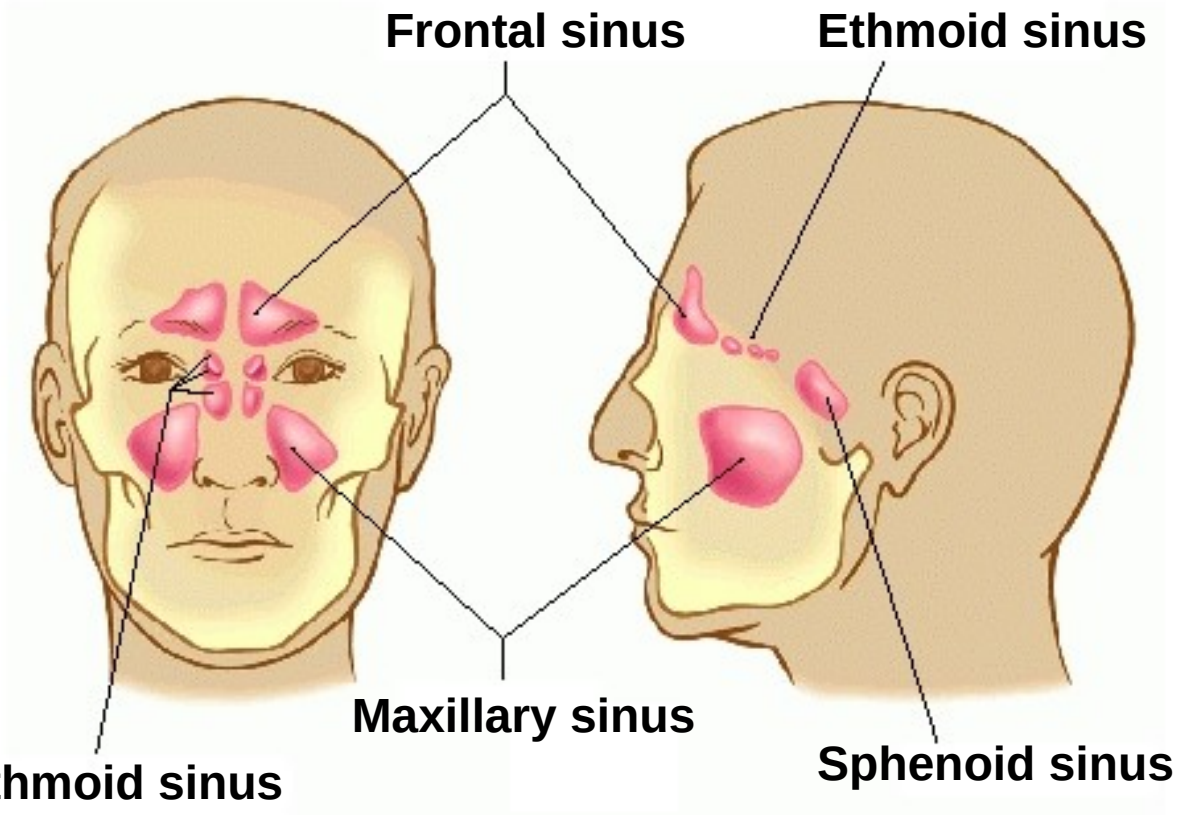


- Open mouth waters
- Maxillary and sphenoid





Identify???


Frontal sinus
Ethmoid sinus
Maxillary sinus
Ethmoid sinus
Sphenoid sinus

THANKS